

Data Science Fundamentals in R — Weekly Schedule

Fall 2026 • OCN682 / MBIO612 • Tuesdays 1:00–2:50 pm, BIL 335

*Schedule subject to change

Week	Date	Readings / Lecture Topic	Quiz	Assignment
1	T 25-Aug	Intro to the class; what is data, data structures R and RStudio refresher / data structures At Home: Reading: Browman and Woo (2018) Collecting and recording data reproducibly, metadata Making data sheets: field vs. analysis		Set up R / RStudio
2	T 1-Sep	Reading: Lowndes et al. (2017); Jennings et al. (2023) What is reproducible research and why does it matter? Tools for reproducible research (GitHub) At Home: R4DS Ch 2, Ch 6 Reproducible scripts and workflow / importing data	1	Set up your file structure and make a script/ Set-up GitHub accounts
3	T 8-Sep	Reading: Healy Ch 1, R4DS Ch 9–11 What makes a good visual? At Home: Wilke Ch 1–5 Visuals part II	2	Practice with ggplot More ggplot (at-home)
4	T 15-Sep	Reading: R4DS Ch 3 Data Wrangling I (group_by, filter, select, summarise, mutate) At Home: R4DS Ch 5 Data Wrangling II (tidyr — pivots, separate, unite)	3	Practice with dplyr Practice with tidyr (at-home)
5	T 22-Sep	Reading: R4DS Ch 17 and 19 Relational data and dates/times (lubridate) At Home: Fun advanced plotting (gganimate, etc.)	4	Practice different types of joins and dates/times
6	T 29-Sep	Reading: R4DS Ch 28, 29	5	Practice with Quarto

Week	Date	Readings / Lecture Topic	Quiz	Assignment
		Intro to Quarto/ Good plot–Bad plot assignment directions At Home: Quarto part II		
7	T 6-Oct	Reading: Wilke Ch 15, Healy Ch 7 Working with spatial data and making maps At Home: Working with spatial data and making maps, part II	6	Make a map
8	T 13-Oct	Reading: RD4S Ch 25 Functional Programming At Home: Launch of the AI-Assisted Community Data Sprint — assign community partner problem and review Ethical & Responsible AI Use Policy	7	Create a function / Start AI-assisted data sprint
9	T 20-Oct	Good plot / Bad plot presentations At Home: Working with factors	8	Final project proposal due
10	T 27-Oct	Reading: R4DS Ch 14; R4DS Ch 15 Working with strings and regex At Home: Work on AI-Assisted Community Data Sprint	9	Working with words
11	T 3-Nov	NO IN-PERSON CLASS — Election Day (UH holiday) Independent work day: continue the AI-Assisted Community Data Sprint	—	AI-Assisted Community Data Sprint: Reprex + AI Developer Log due Nov 10th before class
12	T 10-Nov	Reading: Mastering Shiny Ch 1 Intro to Shiny apps Guided debrief discussion of the Community Data Sprint At Home: Work on final project	10	Make a shiny app

Week	Date	Readings / Lecture Topic	Quiz	Assignment
13	T 17-Nov	Reading: R4DS Ch 26 Iterative data (for loops and map functions) At Home: Work on final project	11	Practice loops, etc.
14	T 24-Nov	Reading: Tidy Modeling with R Models part I and II (many models with broom) Note: <i>No in-person class, Thanksgiving week — online only</i> At Home: Work on final project	12	Practice with lots of models
15	T 1-Dec	Work on Final project	13	
16	T 8-Dec	Final project presentations (group 1)		
17	T 15-Dec (Exam week)	Final project presentations (group 2)		